



FACULTY OF BUSINESS AND IT

MBAI 5600G - Applied Integrative Analytics Capstone Project

Milestones 5

Due: July 7, 2026

Milestone	Description	Due Date
1	Capstone project selection/proposal	May 19
2	Literature review and data source identification	May 26
3	Data collection & Initial EDA	June 9
4	Data Preprocessing & Baseline Modeling	June 23
5	Advanced Modeling & Optimization	July 7
6	Validation & Business Impact Assessment	July 21
7	Capstone project oral presentation	July 27 & August 10
8	Capstone project report	August 10

Important Note on Pace and Deadlines

Project complexity and team progress may vary throughout the term. If your group completes a milestone before its scheduled deadline, you are encouraged to begin working on the next milestone immediately rather than waiting for the due date.

The posted deadlines represent the maximum allowable completion time for each milestone and should not be interpreted as required pacing checkpoints. Early progress is strongly encouraged to allow additional time for refinement, testing, validation, and project improvement.

Milestone 5: Advanced Modeling & Optimization

Milestone 5 is designed to extend the baseline modeling work completed in Milestone 4 by implementing the **novel contribution proposed in Milestone 1** and **conducting systematic model optimization and comparative evaluation**.

At this stage, students move beyond reproducing the baseline methodology presented in the selected journal article and begin introducing their own research contribution, enhancement, or innovation. The goal is to determine whether the proposed modifications improve model performance, interpretability, robustness, scalability, or business value compared to the established baseline benchmark.

The advanced modeling and optimization process should clearly explain:

- How the proposed novel contribution was implemented,
- What advanced models or enhancements were introduced,
- How hyperparameters were optimized,
- How model performance was systematically evaluated,
- How advanced models compare with baseline models,
- What performance improvements were achieved,
- What trade-offs exist between model complexity and performance gains,
- How the results support the project objectives and business problem.

The primary objective of Milestone 5 is to develop, optimize, and evaluate advanced modeling approaches that extend the baseline implementation and demonstrate the value of the proposed research contribution.

1. Novel Contribution Implementation

Students must implement the enhancement, modification, or innovation proposed in Milestone 1. Students must clearly document:

- The proposed contribution from Milestone 1,
- The motivation for the enhancement,
- The theoretical or practical rationale,
- The implementation methodology,
- Integration with the baseline model,
- Expected benefits and improvements.

2. Advanced Model Development

Students must develop and implement one or more advanced models that incorporate the proposed contribution. Students should provide:

- Description of model architecture
- Training procedures,
- Model assumptions,
- Design decisions,
- Implementation details and computational requirements.

3. Hyperparameter Optimization

Students must systematically optimize model performance through structured hyperparameter tuning.

Students must document:

- Parameters selected for tuning,
- Search methodology,
- Evaluation criteria,
- Validation procedures,
- Computational considerations.

4. **Comparative Model Evaluation**

Students must conduct a rigorous comparison between baseline and advanced models.

Students should compare:

- Baseline model(s),
- Enhanced model(s),
- Optimized model(s).

Note: Students should use performance metrics appropriate for their project type. Moreover, Students should provide:

- Performance comparison tables,
- Statistical summaries,
- Visualizations,
- Ranking of models,
- Discussion of observed improvements.

5. **Model Interpretation and Explainability**

Students must analyze and interpret advanced model behavior. Students should examine:

- Feature importance,
- Key predictive drivers,
- Model decision patterns,
- Business interpretation of results,
- Alignment with domain knowledge.

6. **Complexity vs. Performance Trade-off Analysis**

Students must critically evaluate whether performance improvements justify increased model complexity. Students should analyze:

- The types of improvements achieved (e.g., accuracy gains, computational costs, training time, inference time, model interpretability, scalability, maintenance requirements, deployment feasibility..., etc.)
- Whether improvements are statistically or practically significant,
- Situations where simpler models may still be preferred,
- Risks associated with increased complexity,
- Business implications of model selection.

Note: The objective is not simply to achieve the highest performance but to identify the most appropriate model for the project context.

Milestone 5 Deliverables

Students must submit one professionally formatted document (PDF or Word) containing the following sections:

1. Title Page

The title page should include:

- Project Title
- Student Group Name
- Full Names of Team Members

2. Updated Executive Summary (150–250 words)

The updated summary should briefly describe:

- Advanced modeling activities,
- Novel contribution implementation,
- Hyperparameter optimization efforts,
- Comparative performance results,
- Major findings,
- Key trade-offs identified,
- Planned validation activities

3. Novel Contribution Implementation

Students must include:

- Description of the proposed enhancement,
- Research motivation,
- Technical implementation details,
- Integration with baseline methodology,
- Expected benefits.

4. Advanced Model Development

Students must document:

- Advanced algorithms implemented,
- Model architectures,
- Training procedures,
- Configuration settings,
- Computational considerations.

5. Hyperparameter Optimization Report

Students must provide:

- Search strategy used,
- Parameters tuned,
- Optimization workflow.
- Parameter search space,
- Best parameter combinations,
- Performance improvements.

6. Comparative Model Evaluation

This section must include:

Performance Results

Students must provide:

- Baseline model results,
- Advanced model results,
- Optimized model results,
- Performance comparison tables,
- Evaluation metrics,
- Visualizations.

Comparative Analysis

Students should explain:

- Performance differences,
- Improvement achieved,
- Strengths and weaknesses of each approach,
- Statistical or practical significance of results.

7. Model Interpretation and Insights

Students must include:

- Feature importance analysis,
- Explainability results,
- Business interpretation of findings,
- Insights generated from advanced models.

8. Complexity vs. Performance Trade-off Assessment

Students must discuss:

- Computational costs,
- Model complexity,
- Scalability considerations,
- Interpretability implications,
- Business deployment considerations,
- Overall recommendation.

9. References (IEEE Style)

Students must use proper IEEE citation formatting consistently throughout the document.

Example References

[1] J. Smith and A. Kumar, "Explainable ensemble learning for customer churn prediction," *IEEE Access*, vol. 13, pp. 11234–11249, 2025.

[2] R. Chen et al., "Comparative analysis of deep learning and traditional machine learning methods for financial forecasting," *Expert Systems with Applications*, vol. 245, 2026.

Submission Guideline

Please submit one **Advanced Modeling & Optimization Report** per group through Canvas no later than **July 7, 2026, at 11:59 PM**.